Symbiosis Institute of Technology, Nagpur

BIG DATA ANALYTICS MINI PROJECT REPORT

for

Bachelor of Technology

in

Computer Science and Engineering

Batch 2023-27

On

Business Intelligence Dashboard using Big Data Analytics

Telecom Customer Churn Analysis using Power BI, Python and Machine Learning

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Institute Guide

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1. Introduction & Problem Statement

Telecom is a subscription-driven business, where sustained revenue depends far more on retaining existing customers than on acquiring new ones. Despite this, a significant share of subscribers cancel their service every year, and without a systematic way to identify who is leaving and why, retention teams are left reacting after the fact instead of intervening in time.

This project was undertaken with the following objectives:

- Analyze historical customer data to uncover the patterns behind churn.
- Identify the contract types, services and pricing tiers linked to higher churn risk.
- Build a machine-learning model that predicts which customers are likely to leave.
- Translate these predictions into a risk-ranked list that retention teams can act upon.

With over one in four customers churning, even small, well-targeted retention actions can translate into meaningful revenue protection, making this analysis directly relevant to the business.

Industry experience across subscription businesses shows that retaining an existing customer is consistently cheaper than acquiring a new one, which is why churn prediction has become a core analytics use case for telecom operators. Framing churn as a forward-looking prediction problem, rather than a lagging report of who has already left, allows retention teams to move from reactive win-back offers to proactive, targeted outreach while the customer relationship can still be saved.

2. Dataset Overview

The project uses the publicly available Telco Customer Churn dataset from Kaggle, which contains customer demographics, account information, subscribed services and churn status.

7,043 Customer Records	**21** Attributes per Customer	**26.54%** Overall Churn Rate	**1,407** Customers Scored by ML Model

Source: https://www.kaggle.com/datasets/blastchar/telco-customer-churn/data .

After cleaning, the working dataset contained 7,032 customers, which was subsequently split into four purpose-built tables for use in Power BI.

Before loading into Power BI, the raw Kaggle export was preprocessed in Python: column names were standardized, TotalCharges was converted to a numeric type, the churn outcome was preserved in its numeric (0/1) form, and the customer identifier was kept consistent across every output so the tables could later be related on customerID. The cleaned data was then split into four purpose-built CSV files, each representing a distinct business entity rather than one flat table:

File	Purpose	Rows
Customer_Details.csv	Customer demographic, account and billing information (gender, senior-citizen flag, partner/dependents, tenure, contract, paperless billing, payment method, monthly and total charges).	7,032
Customer_Services.csv	Phone and internet service subscriptions, including add-ons such as online security, online backup, device protection, tech support and streaming services.	7,032
Churn_Analysis.csv	Customer ID mapped to the actual churn outcome (0 = retained, 1 = churned).	7,032
Telcom_ML.csv	Pre-scored ML test set used to power the predictive dashboard; already cleaned in Python before being loaded into Power BI.	1,407

Table 2.1 — The four purpose-built tables produced during Python preprocessing

Customer_Details carries eleven attributes per customer — customerID, gender, SeniorCitizen, Partner, Dependents, tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges and TotalCharges — while Customer_Services tracks the specific services each customer has subscribed to:

- PhoneService and MultipleLines
- InternetService (DSL, Fiber optic or No internet service)
- OnlineSecurity, OnlineBackup and DeviceProtection
- TechSupport, StreamingTV and StreamingMovies

Deliberately, values such as "No phone service" and "No internet service" were kept as their own categories rather than being collapsed into a plain "No", since they carry meaningful information about a customer's overall service configuration that a simple Yes/No flag would lose.

3. Project Methodology

The project follows a descriptive → diagnostic → predictive analytics journey, moving from raw data to a live, interactive Power BI report in four stages:

- Prepare in Python: The Telco Customer Churn dataset (7,032 customers) was cleaned in Python — columns were standardized, TotalCharges was converted to numeric, and the data was split into four purpose-built tables, including a separate 1,407-row ML scoring set.
- Model in Power BI: The files were connected via a single Folder query (not Combine), preserving each table's individual structure. Text columns were trimmed and cleaned, data types were verified, and customerID relationships were built between Customer_Details, Customer_Services and Churn_Analysis.
- Predict with ML: Five classification models — Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest and Support Vector Machine (SVM) — were trained and compared on Accuracy, Precision, Recall, F1-score and ROC-AUC. KNN was selected as the best-performing model.
- Visualize in 3 Pages: The results were delivered as three linked Power BI report pages — Executive Overview, Customer & Service Analysis, and ML Prediction & Risk — each answering a progressively deeper business question.

4. Data Transformation in Power BI (Power Query)

Before any modelling or visualization took place, every source table was connected, profiled and cleaned inside Power Query. The screenshots in this section are captured directly from the project's own Power BI Desktop session and illustrate the exact transformation steps that were carried out.

Step 1 — Connect the source folder

Rather than importing each CSV file individually, the entire folder containing the four source files was connected using Get Data → Folder → Transform Data. The folder path used was C:\Users\sande\OneDrive\Desktop\Telecom.

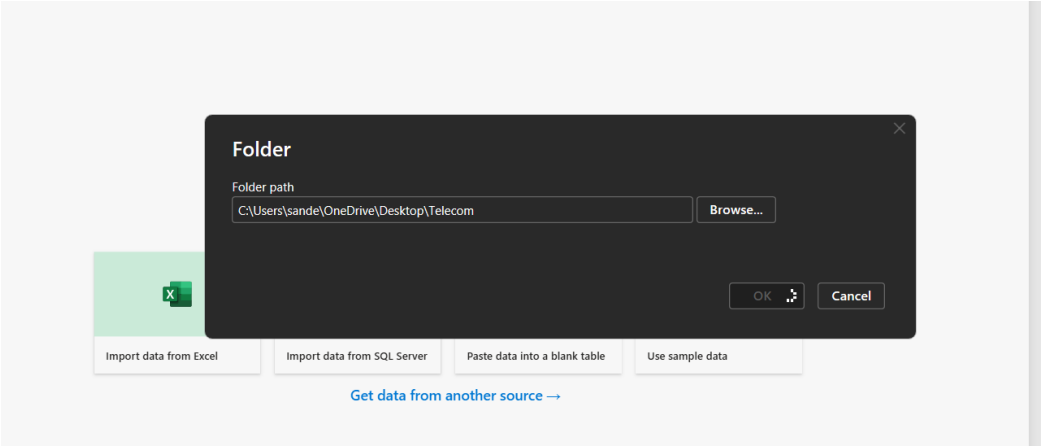


Figure 4.1 — Connecting to the source folder via Get Data → Folder

The folder contained four CSV files — Churn_Analysis, Customer_Details, Customer_Services and Telecom_ML. It was deliberately decided not to use the Combine Files option, since these CSVs have different column structures and represent different business tables. Instead, each file was kept as a separate Power Query query, and Transform Data was selected to open the Power Query Editor for individual cleaning.

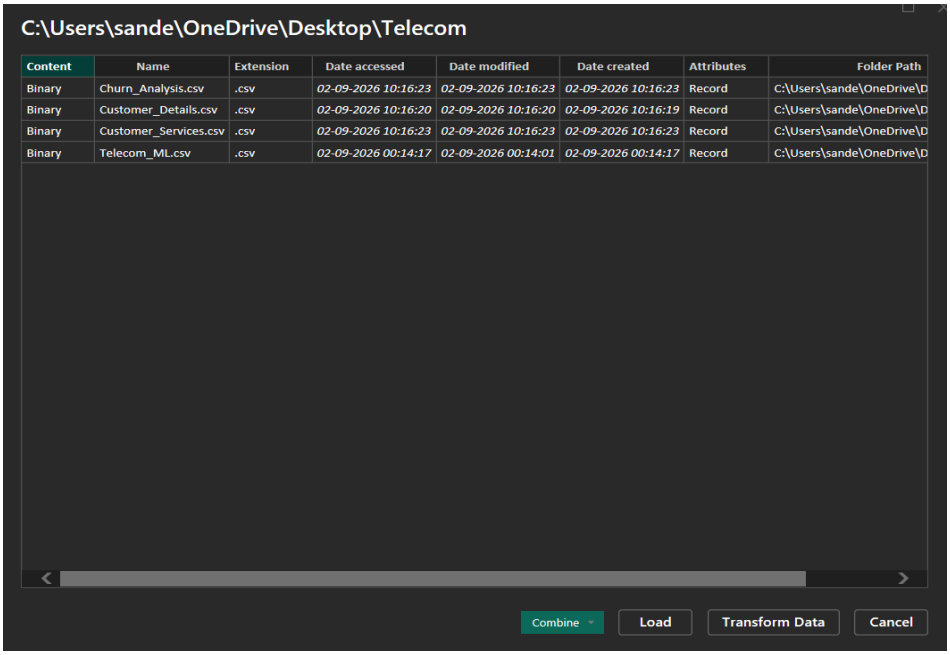


Figure 4.2 — The four source CSV files identified inside the connected folder

Step 2 — Profile every column

With all four queries loaded — Churn_Analysis, Customer_Details, Customer_Services and Telcom_ML — each table was inspected individually, starting with Customer_Details. Column quality, Column distribution and Column profile were enabled from the View tab, with profiling based on the entire dataset rather than only the preview rows. This made it possible to identify missing values, errors, incorrect data types and duplicate or invalid entries across every column at a glance.



Figure 4.3 — The four Power Query tables and the applied transform-column-types step

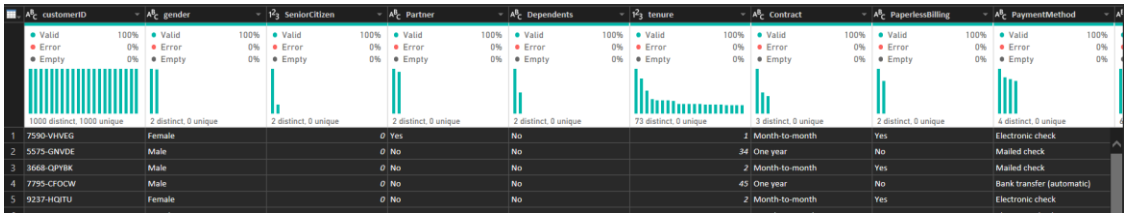


Figure 4.4 — Column quality, distribution and profile view across Customer_Details columns

Step 3 — Clean text columns

The categorical text columns — gender, Partner, Dependents, Contract, PaperlessBilling and PaymentMethod — were selected together and cleaned in two passes: Transform → Format → Trim, followed by Transform → Format → Clean. This removed unnecessary leading/trailing spaces and unwanted non-printing characters that are common in raw exported data.

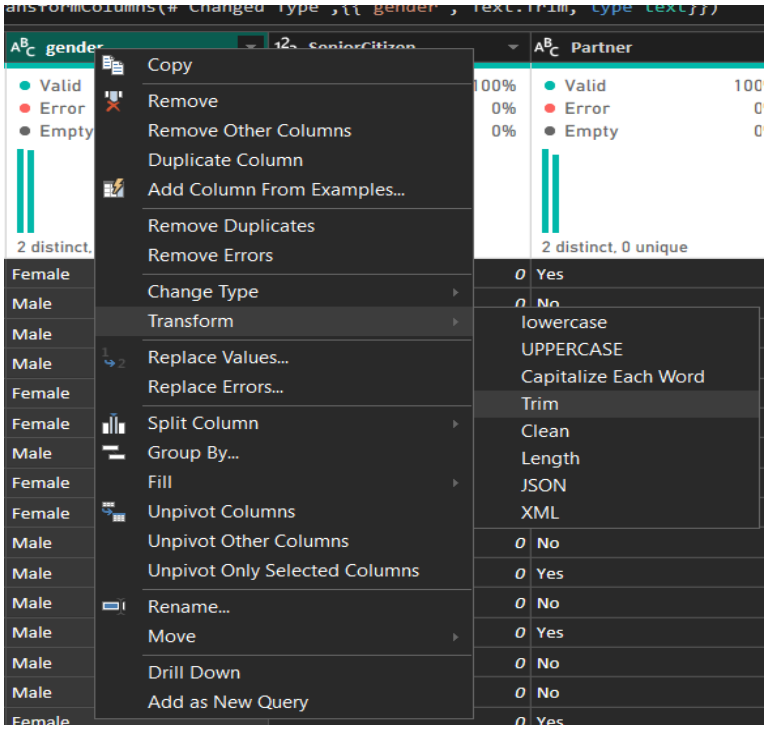


Figure 4.5 — Applying Trim and Clean transformations to categorical text columns

Step 4 — Fix data types

Each column's data type was then explicitly set and verified: customerID, gender, Partner, Dependents, Contract, PaperlessBilling and PaymentMethod were kept as Text; SeniorCitizen and tenure were set to Whole Number; and MonthlyCharges and TotalCharges were set to Decimal Number. Categorical columns were deliberately left as text and not converted to numeric types, to preserve their meaning.

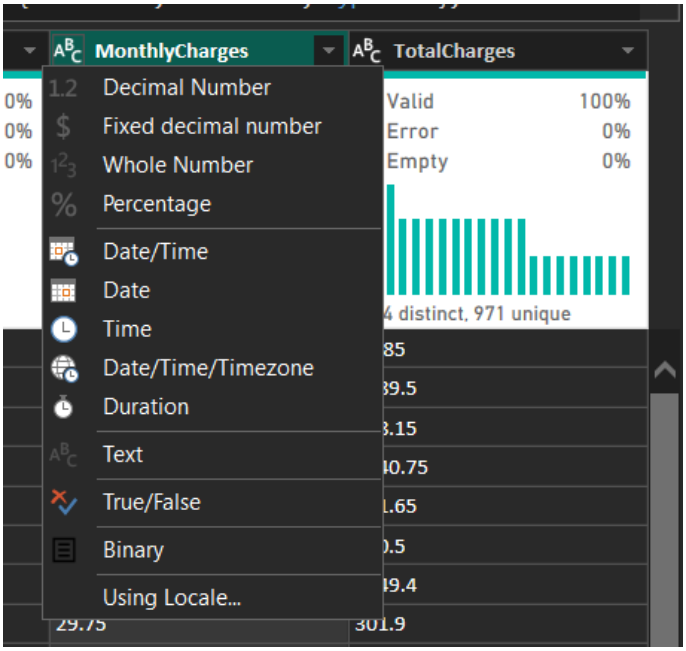


Figure 4.6 — Setting the correct data type (Decimal Number) for MonthlyCharges and TotalCharges

Step 5 — Check for duplicates

Before finalizing the Customer_Details table, the customerID column was checked for uniqueness using the column profile (based on the entire dataset, not just the preview). The profile confirmed 7,043 distinct and 7,043 unique values, meaning every customer appears exactly once and no duplicate removal was required.

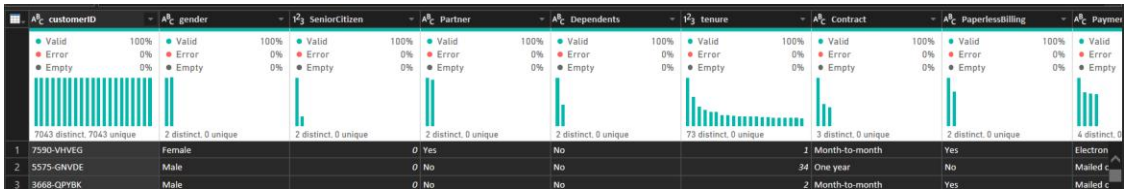


Figure 4.7 — Verifying customerID uniqueness across the full dataset (7,043 distinct / 7,043 unique)

For Customer_Services, the same Trim/Clean process was discussed, and a deliberate decision was made not to merge “No phone service” / “No internet service” into a simple “No”, since they carry meaningful information about each customer's service configuration. For Churn_Analysis, the Churn column was confirmed to be represented numerically (0 = No Churn / Retained, 1 = Churned) and was left unchanged. The Telcom_ML table had already been cleaned in Python, so no further transformation was applied to it in Power Query — only the imported data types were verified before loading.

Once every table was verified, Close & Apply was used to load all four cleaned tables — Customer_Details, Customer_Services, Churn_Analysis and Telcom_ML — into the Power BI data model, ready for relationship building.

5. Power BI Data Model (Model View)

With the cleaned tables loaded, the data model was built inside Power BI's Model View. Four tables connect through a single shared key, customerID, keeping demographic, service, churn-outcome and ML-prediction data cleanly separated into distinct tables while remaining fully queryable together.

- Customer_Details — demographics, contract and billing information (7,032 rows).
- Customer_Services — phone, internet and add-on service subscriptions.
- Churn_Analysis — the actual churn outcome for each customer (0 / 1).
- Telecom_ML — the 1,407-row KNN prediction and churn-probability set.

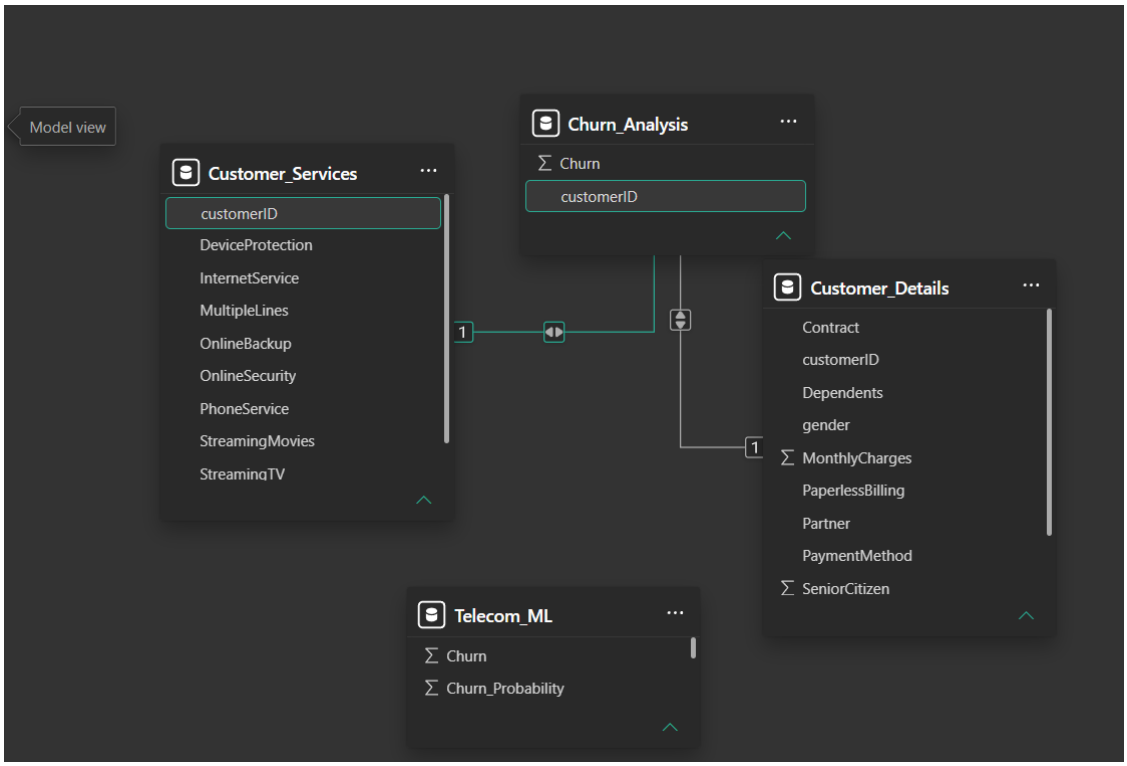


Figure 5.1 — Actual Power BI Model View showing relationships built on customerID

For the three 7,032-row tables, one-to-one (1:1) relationships with single-direction cross filtering were created:

Customer_Details[customerID] → Customer_Services[customerID], and Customer_Details[customerID] → Churn_Analysis[customerID]. Each relationship was created by dragging the customerID field between tables and explicitly setting the cardinality to One-to-one and the cross-filter direction to Single, with “Make this relationship active” enabled. The Telcom_ML table was deliberately kept unconnected to the main model at this stage, since its structure and purpose needed to be verified first to avoid producing misleading dashboard results; it is instead used directly to power the predictive-analysis page.

Each relationship was created by explicitly dragging the customerID field between tables in Model View and setting the cardinality and cross-filter direction by hand, rather than relying on Power BI's autodetect feature, to make sure the model behaved predictably:

From	To	Cardinality	Cross Filter
Customer_Details[customerID]	Customer_Services[customerID]	1 : 1	Single
Customer_Details[customerID]	Churn_Analysis[customerID]	1 : 1	Single

Table 5.1 — Relationships defined in the Power BI data model

6. Power BI Dashboards

The final report was delivered as three linked Power BI dashboard pages, each going one level deeper than the last — moving from a descriptive view of what is happening, to a diagnostic view of why it is happening, to a predictive view of who is likely to churn next.

6.1 Dashboard 1 — Executive Overview (Descriptive)

The first dashboard, Telco Customer Churn Analytics – Executive Overview, answers one central question: what is happening with customer churn? The KPI cards give an overall snapshot of the customer base — 7,043 total customers, of which 1,869 have churned, resulting in an overall churn rate of 26.54%, while 5,174 customers remain active.

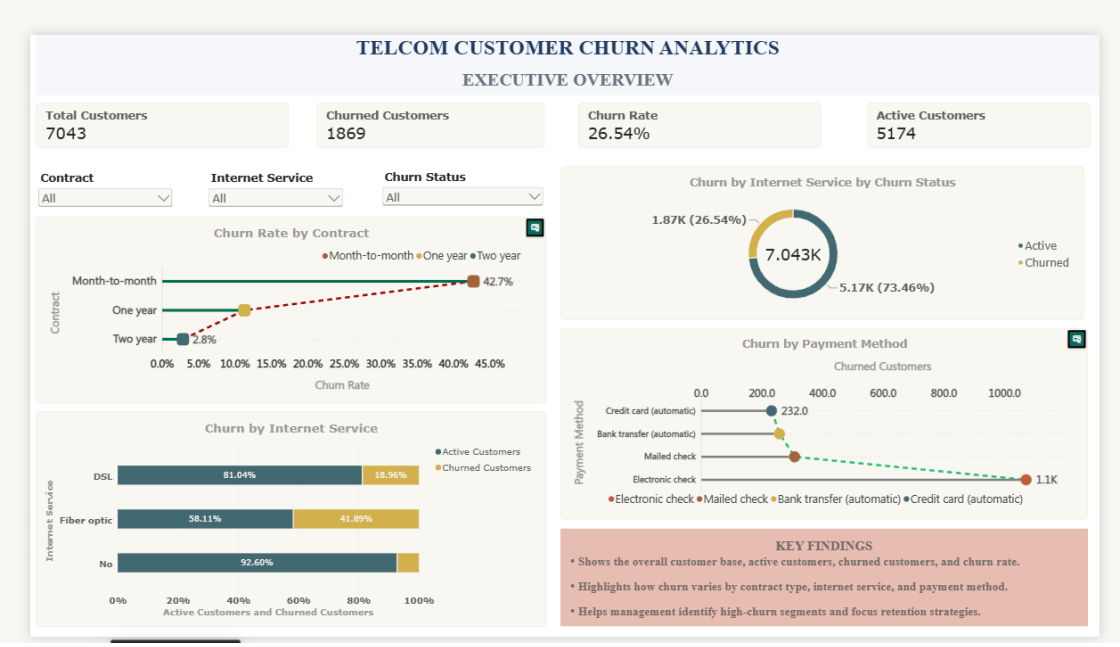


Figure 6.1 — Dashboard 1: Executive Overview

The Churn Rate by Contract lollipop chart shows that month-to-month customers have by far the highest churn rate at 42.7%, compared to just 2.8% for two-year contracts — customers without a long-term contract are considerably more likely to leave. The Customer Churn Distribution donut chart gives a simple overall view of the proportion of active versus churned customers. The Churn by Internet Service chart shows that Fiber Optic customers have a notably higher churn proportion of 41.89%, while the Churned Customers by Payment Method chart shows that Electronic check is associated with the highest number of churned customers.

Key Findings — Dashboard 1

- Shows the overall customer base, active customers, churned customers, and churn rate.
- Highlights how churn varies by contract type, internet service, and payment method.
- Helps management identify high-churn segments and focus retention strategies.

6.2 Dashboard 2 — Customer & Service Analysis (Diagnostic)

The second dashboard takes the analysis one step further and answers: what factors are associated with customer churn? The Churn by Gender chart shows very similar churn rates for males (26.16%) and females (26.92%), indicating gender is not a major differentiating factor.

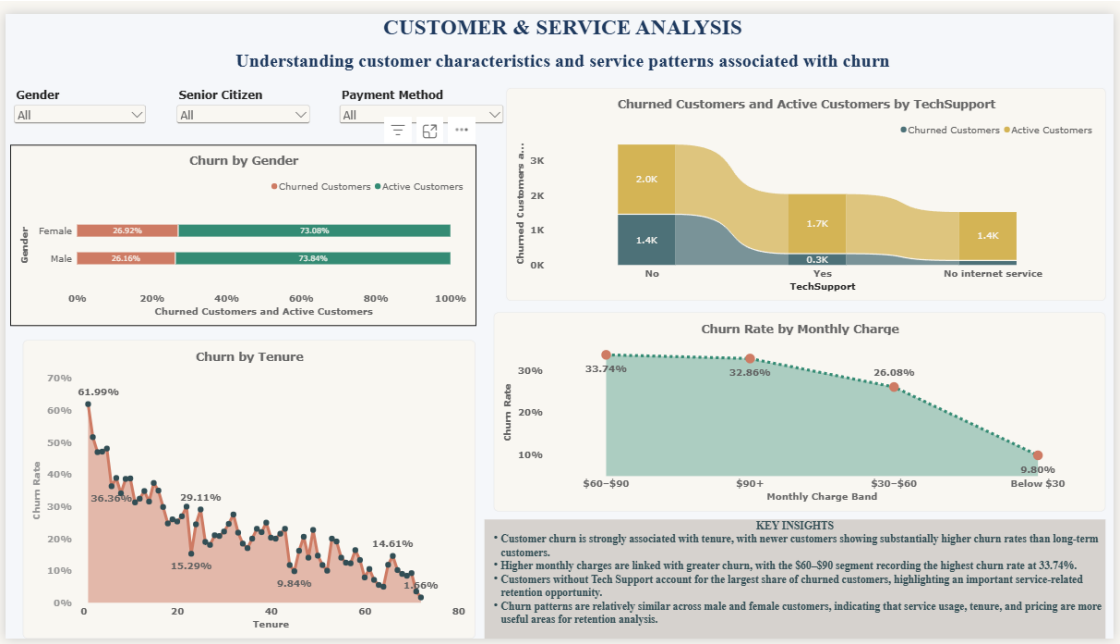


Figure 6.2 — Dashboard 2: Customer & Service Analysis

The Churned and Active Customers by Tech Support chart shows that customers with no Tech Support form the largest group and also account for a large share of churned customers. The Churn by Tenure line & area chart shows that newer customers churn at a much higher rate (61.99% in the first period) and churn steadily decreases as tenure increases, making the early customer relationship a critical retention period. The Churn Rate by Monthly Charge chart shows the $60–$90 band has the highest churn rate at 33.74%, followed by the $90+ band at 32.86%, while customers paying below $30 churn at only 9.80%.

Key Insights — Dashboard 2

- Customer churn is strongly associated with tenure, with newer customers showing substantially higher churn rates than long-term customers.
- Higher monthly charges are linked with greater churn, with the $60–$90 segment recording the highest churn rate at 33.74%.
- Customers without Tech Support account for the largest share of churned customers, highlighting an important service-related retention opportunity.
- Churn patterns are relatively similar across male and female customers, indicating that service usage, tenure, and pricing are more useful areas for retention analysis.

6.3 Dashboard 3 — ML Prediction & Risk Analysis (Predictive)

The third dashboard moves the project from descriptive and diagnostic analysis into predictive analytics, answering the final question: who is likely to churn? For the predictive analysis, five classification models — Logistic Regression, KNN, Decision Tree, Random Forest and SVM — were evaluated using Accuracy, Precision, Recall, F1-score and ROC-AUC. KNN was selected as the best-performing model and is used throughout this dashboard, which analyzes 1,407 customers.

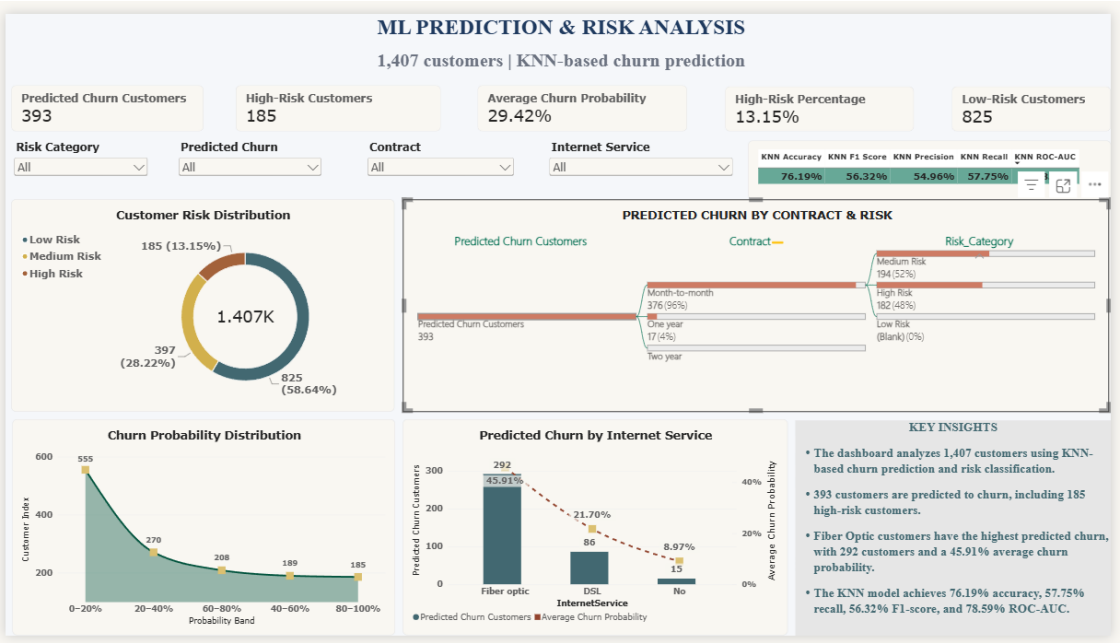


Figure 6.3 — Dashboard 3: ML Prediction & Risk Analysis

The KPI cards show that the model predicts 393 customers as likely to churn, including 185 high-risk customers, at an average churn probability of 29.42% and a high-risk percentage of 13.15%; 825 customers are classified as low risk. The Customer Risk Distribution donut chart breaks this down into Low Risk (825, 58.64%), Medium Risk (397, 28.22%) and High Risk (185, 13.15%). The Predicted Churn by Contract & Risk decomposition tree shows that of the 393 predicted churners, 376 (about 96%) are on month-to-month contracts; within that month-to-month segment alone, 194 customers are classified medium risk and 182 high risk, confirming that this single contract type drives almost the entire predicted-churn population. The Churn Probability Distribution chart shows the largest customer group (555) falls in the 0–20% probability band, followed by 270 customers in the 20–40% band. The Predicted Churn by Internet Service chart shows Fiber

Optic has the highest predicted churn — 292 customers at a 45.91% average probability — compared with 86 predicted churners on DSL at a 21.70% average probability and just 15 predicted churners with no internet service at an 8.97% average probability. The KNN Model Performance cards report 76.19% accuracy, 54.96% precision, 57.75% recall, 56.32% F1-score and 78.59% ROC-AUC.

Key Insights — Dashboard 3

- 393 customers are predicted to churn, 185 of them classified as high risk.
- 96% of predicted churn customers are on month-to-month contracts.
- Fiber Optic customers have the highest predicted churn, with an average churn probability of 45.91%.
- The KNN model achieves 76.19% accuracy, 57.75% recall, 56.32% F1-score, and 78.59% ROC-AUC.

7. Predictive Modelling — Model Comparison

Five classification algorithms were trained and compared on the same set of customer features, with the final model chosen on balanced performance rather than accuracy alone: Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest and Support Vector Machine (SVM). Each model was evaluated on Accuracy, Precision, Recall, F1-score and ROC-AUC.

76.19% Accuracy	**54.96%** Precision	**57.75%** Recall	**56.32%** F1-Score	**78.59%** ROC-AUC

K-Nearest Neighbors (KNN) was selected as the best-performing model on this dataset, and its predictions power the third dashboard described above.

8. Synthesis — Key Findings End to End

- Scale: 26.54% of customers have churned — a material, recurring revenue loss.
- Segment: Month-to-month contracts, Fiber Optic service and the absence of Tech Support consistently over-index on churn.
- Lifecycle: Risk is front-loaded — new customers churn far more than long-tenured ones, and higher monthly bills raise risk further.
- Prediction: The KNN model (76.19% accuracy, 78.59% ROC-AUC) identifies 393 likely churners, 185 of them high-risk.

9. Recommendations

- Prioritize the 185 high-risk customers: route the ML-flagged high-risk list to retention teams for proactive, personalized outreach this quarter.
- Incentivize contract upgrades: offer targeted discounts or perks to move month-to-month customers onto 1- or 2-year plans, where churn drops to just 2.8%.
- Bundle Tech Support proactively: promote or bundle Tech Support with Fiber Optic plans, since both are strongly linked to higher churn when absent.
- Watch the first 90 days closely: build an early-tenure onboarding and check-in program, since churn risk is highest in a customer's first months.
- Segment retention spend by risk tier: since Fiber Optic customers carry a 45.91% average predicted churn probability versus 21.70% for DSL and 8.97% for customers with no internet service, retention budget should be weighted toward Fiber Optic and month-to-month accounts rather than spread evenly across the base.

10. Conclusion

This project combined a carefully cleaned and modelled Power BI data foundation with descriptive, diagnostic and predictive analytics to build a complete picture of customer churn. Dashboard 1 established what is happening with churn, Dashboard 2 explained what factors are associated with it, and Dashboard 3 used a KNN-based machine learning model to predict who is likely to churn next and how risky those customers are. Together, the three linked dashboards — supported by a clean data-transformation pipeline in Power Query and a well-structured relational data model — give retention teams a practical, risk-ranked tool for reducing customer churn.